

Teacher-Informed Tactile Sensing Upgrade for Child–Robot Physical Interaction: A Comparison of Velostat and FBG Sensors

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Abstract—Social robots designed to engage with children require additional co-designed features that reflect authentic interactions. Physical human robot interaction (pHRI) in this application requires tactile sensors that are reliable and safe for natural children’s touch. This work reports on the design and assessment of two tactile sensing systems—one using Velostat and the other based on optical sensing—developed in line with teacher feedback to enhance the integration of the CASTOR robot into educational activities. Both sensors were characterized under controlled forces representative of human touch. The Velostat sensor exhibited a nonlinear force response, higher spatial resolution, and lower hysteresis, while the optical sensor demonstrated a highly linear response and greater sensitivity. The sensing systems were subsequently evaluated with 5 participants (25.73 ± 8.32 years) in a hand-shaking pHRI task. A heatmap-based analysis revealed that both sensors effectively captured the temporal and spatial distribution of contact forces, while also highlighting qualitative differences in localized forces among participants, with a maximum exerted force of 1.30 N. Overall, the results indicate that the Velostat-based sensor is advantageous for the educational scenario, as it provides detailed contact localization, exhibits low hysteresis, is more robust, and is replicable.

I. INTRODUCTION

Social robots are increasingly deployed in healthcare, assistance, education, and service contexts, where they engage in close and prolonged interactions with users [1]. As an essential consideration, custom and adaptable activities and functionalities in social robotics have demonstrated improved acceptance and an increased intention to use among specific communities, such as in autism [2] and education [3].

In addition, when interacting with children who tend to engage in frequent and spontaneous physical contact, the user’s perception of the robot can be strongly influenced by the quality and responsiveness of sensing systems [4]. The behaviour of robots and the free touch can generate engagement and improve attitudes toward the robot in therapeutic and educational settings [4], [5]. For such “in the wild” scenarios, reliable sensing capabilities are essential to support the robot’s decision-making, reduce behavioral

errors, and ensure safe and natural physical human–robot interaction (pHRI) [6].

Participatory design (PD) is increasingly recognised as an effective approach for developing technologies that reflect users’ real needs and preferences [2]. In social robotics, PD has involved autistic children and teachers in co-designing educational robots [3]. However, most educational applications remain focused on supporting children with ASD individually rather than fostering shared interactions and inclusive activities within mainstream classrooms [7].

In this sense, the CASTOR Project is a collaborative open-source and child-oriented social robot designed for therapeutic settings [8]. It has functionalities to give social cues to motivate and increase the children’s attention to the therapeutic activities, it is adaptable to the children’s conditions [9], and currently is being adapted through co-design to its implementation in a primary School in the UK in educational contexts. For this, 12 staff members from a UK primary school participated in semi-structured interviews to identify desirable improvements to the CASTOR robot for use with the Year 2 class (7-8-year-olds). The interview session was organized in two parts: the presentation of the robot and its functionalities; and the interview to generate feedback about the teachers’ first impressions of the CASTOR robot, the potential uses in the classrooms, usability and teachers’ perspectives, concerns and risks, and improvements to the robot to be used in the school. All participants had experience teaching children aged 4–11 and were familiar with everyday classroom routines and learning requirements, and none had prior experience using socially assistive robots in their teaching practice. This work will not present the entire feedback analysis; instead, it will focus on the touch-sensitive arms, which resulted from a teacher-informed upgrade to the CASTOR robot. This was prioritized to enhance interactivity, safety, and generate educationally relevant activities in the primary school.

Considering the need for a sensor to achieve the touch awareness function in the CASTOR robot, and the different kinds of child-robot pHRI that can be presented (rough to soft, such as hits, handshaking, and hugs) [10], [11]; the mechanical and functional characteristics of the sensing system must meet the requirements to control the robots’ reactions [12]. The robustness, sensitivity, hysteresis, repeatability, spatial resolution, and cost are among the attributes considered in determining whether a sensing technology is suitable for an application [13]. For touching detection in robotic applications, a wide variety of sensing principles have

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been explored, including capacitive sensing [14]; resistive sensing [15]; and piezoresistive sensing [16]. More recently, optical sensing approaches have been used, offering new opportunities for improved resolution and sensitivity [17].

This work contributes to the state of the art in social robots' adaptability to users' needs by (1) developing two distinct touch-sensing systems, (2) characterizing them to compare mechanical and functional performance, and (3) integrating both sensors into the robot's hand to perform an initial evaluation of pHRI. The objective is to determine which of the two developed sensors is better suited for the application scenario with children, as a functionality obtained from the community feedback in the interview phase; the framework is illustrated in Figure 1.

II. RELATED WORK

This section reviews prior work on tactile sensing technologies applied to social robots and pHRI, with a focus on velostat-based and optical tactile sensors.

A. Touch sensors in Social Robots

Tactile sensing has become a fundamental component for enabling safe and natural physical interaction between humans and social robots [15]. Early efforts in social HRI equipped robots with capacitive skins or resistive patches to detect touches on large body areas, primarily to support rough interactions [14]. These approaches demonstrated the importance of responsiveness and localization accuracy for engaging users [18]. Recent tactile systems for social robots have explored multi-modal sensing, compliant skins, and distributed arrays designed to interpret gestures [19], [20]. While these systems have shown promise for improving the detection, many remain expensive or difficult to manufacture for long-term deployment. This motivates the exploration of alternative low-cost materials and sensing principles, such as piezoresistive films and optical fibers, to achieve tactile sensing tailored to social pHRI.

B. Velostat-based tactile sensors

Velostat is a carbon-loaded polyolefin material with piezoresistive properties that has been widely adopted in low-cost tactile sensors due to its flexibility, ease of fabrication, and suitability for large-area skins [21], [22]. Several works have advanced the design and characterization of Velostat sensors for robotics applications [13]. The material's fundamental properties, such as hysteresis, nonlinearity, and response time, were systematically evaluated in [21], providing a reference for calibration and highlighting limitations, such as cross-talk. Velostat has been implemented in different forms, including sandwich-structured 3D force sensors for electronic skins and glove-like arrays [16], interdigital sensors for contact force estimation in robot grippers [22], and matrix-based systems for touch localization, such as the 27×27 array proposed in [23].

Large-area applications have also been demonstrated, including 32×32 matrices for foot-pressure mapping [19], showing scalability for full-body sensors. Design enhancements such as stacking multiple Velostat layers have been

shown to improve accuracy and reduce noise [24], while sensor architectures for fingertips have targeted improved linearity and reduced hysteresis for manipulation tasks [13]. Although Velostat remains attractive for pHRI due to its low cost and adaptability, challenges related to cross-talk and hysteresis still limit its use in social robots.

C. Optical-based tactile sensors

Optical sensors in touch sensing have been useful for robotic devices in assistance. Particularly, fiber Bragg grating (FBG) sensors have emerged as an alternative due to their high sensitivity and precision, immunity to electromagnetic interference, fatigue resistance, and multiplexing capabilities [25]. Flexible or textile-based optical sensors have been developed for deformation and force detection. For example, [26] introduced a bionic skin using a double-layer FBG array capable of detecting pressure, slip, and temperature.

The sensitivity of the FBG sensors in detecting touch and force variations is strengthened by their application in various robotic domains to develop more intelligent systems, including soft robotic grippers [27] and HRI systems [28]. In this sense, D'Abbraccio *et al.* integrated an artificial skin into the forearm of a collaborative robot [29], mapping the interactions and enhancing intuitive behaviors with the users. Despite the great potential of this technology, few implementations have been reported with social robots; for this reason, previous work has tested the FBG sensors concept in the hands of the CASTOR robot [30], [31]. Moreover, the fabrication complexity, the need for specialized interrogation hardware, and integration challenges in soft child-oriented robot platforms lead to a comparison of the developed sensors with other sensing principles. Although Velostat-based sensors offer low-cost scalability, they suffer from mechanical and electrical variability, for which comparative studies are needed to evaluate these sensing approaches under conditions representative of social-robot pHRI. This motivates the present work, which develops, characterizes, and compares two tactile sensing systems integrated into the CASTOR robot's hand. This fulfills the teachers' requirement to make the robot reactive to children's touch, to be implemented in education.

III. METHODOLOGY

This section describes the sensors' development, the calibration, and the assessment procedure in a pHRI by integrating them into the robot's hand.

A. Sensor matrix development

1) *Velostat*: The Velostat-based sensor was built in a rectangle of 110 × 100 mm structure composed of four layers, arranged to allow the formation of 36 sensitive points distributed in a 6 × 6 array, as can be seen in Figure 1b. The top layer consists of insulating tape (3M, USA), applied to ensure electrical insulation of the sensor surface and to secure the 12.5 mm wide copper tapes, which were positioned with the metallic side facing the Velostat, since it was observed that the opposite side reduced the sensor's sensitivity. In the next layer, six vertically oriented copper tape traces were

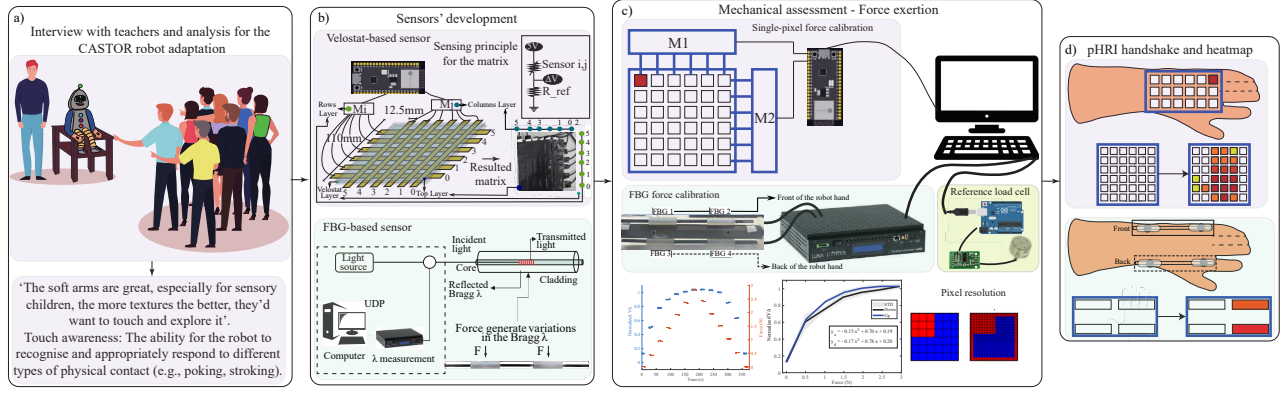


Fig. 1. Framework for the contribution stages of this work to identify the best-suited sensor for pHRI with a social robot's hand. a) Interview process about the CASTOR social robot in education with teachers. b) Sensors' development. c) The mechanical evaluation performed by calibrating the sensors with a reference load cell. d) Heatmap analysis of a handshaking interaction comparing both types of sensors.

placed, corresponding to the system's columns. In the central region, a VelostatTM piezoresistive film (3M, USA) sheet was added, and under it six additional copper traces were placed horizontally, forming the matrix rows. The last layer of the structure was also sealed with insulating tape.

2) *FBG*: The proposed FBGs were fabricated using a UV laser with the phase-mask technique. This process involves inscribing the desired periodic refractive index modulation along the core of a single-mode optical fiber [32]. The UV laser employed (CNI DPS-266, China) operated at 266 nm and emitted pulses of 8 ns at a repetition rate of 100 Hz [33]. The alignment ensured uniform exposure, while the laser exposure time was controlled with an optical sensing interrogator SM125 (Micron Optics, Inc.).

The inscribed FBGs use 3D-printed guides made of Polylactic acid (PLA) for their integration into the robot hand. To enhance its mechanical resistance, the sensitive areas of the proposed network were covered with a silicone layer (EcoFlex, Polisil, Brazil), also increasing the sensing area and improving the sensitivity. This sensor replicates a previously validated design using two Fiber Bragg Gratings distributed across two optical channels, obtaining a 2 x 2 matrix [31]. The matrices were different due to the complexity in multiplexing multiple sensors within a short fiber segment with the UV mask method, and the channels that depend on the optical interrogator. FBG1 and FBG2 were located at the front, while FBG3 and FBG4 were placed at the back. Finally, the λ_{Bragg} for FBG1-and-3 was 1548.3 nm, and for FBG2-and-4 was 1562.6 nm.

B. Sensor calibration

The sensors were characterized with quasi-static loading: Firstly, the sensors were subjected to a progressive load from 0-5 N until it reached saturation [31], which was 3 N for Velostat, and the FBG sensor did not reach saturation in the progressive load test. For which, the Velostat calibration was from 0-3 N in 0.5 N steps and FBGs from 0-5 N in 1 N steps. Each step was held for 15 s, and the full up-down calibration was conducted for every single pixel in

both systems, 36 in the Velostat matrix and four in the FBG matrix. Calibration models (linear, quadratic, and power-law) were fitted by least squares and compared, and the sensitivity, linearity error, hysteresis, repeatability, and resolution were reported. The setup for the calibration of both sensors is shown in Figure 2.

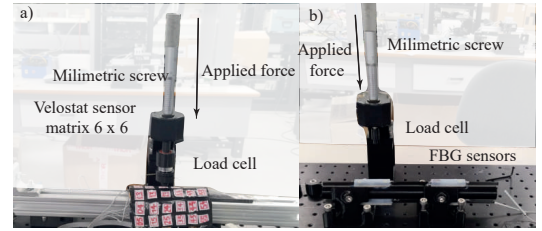


Fig. 2. Calibration setup for single pixel force characterization. a) Velostat, and b) FBG.

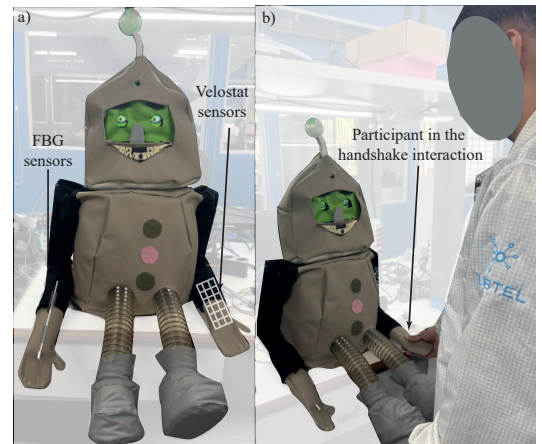


Fig. 3. Setup for the pHRI application. a) Participant's view of the robot and the sensors' placement representation in both hands and forearms of the robot. b) Example of a hand interaction with the robot.

C. Participants

Hand-shaking interaction: Five adult participants (25.73 $\pm$ 8.32 years; 2 female, 3 male) were recruited for this

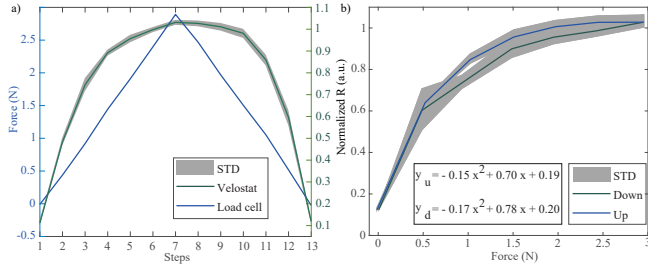


Fig. 4. Resulted characterization for force steps. a) Velostat comparison with the reference load cell, b) Velostat regression to the applied forces.

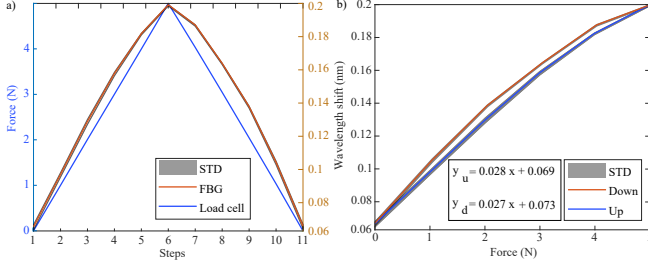


Fig. 5. Resulted characterization for force steps. a) FBG comparison with the reference load cell, and b) FBG regression to the applied forces.

preliminary study. All participants reported normal upper-limb motor function and no sensory impairments. Inclusion criteria required being over 18 years of age, having no prior negative experiences with robots, and those who felt uncomfortable with physical contact with robots were excluded. This study was approved by the ethics committee of the Federal University of Espirito Santo (85461424.7.0000.5542), followed institutional ethical guidelines, and all participants provided informed consent before participation.

D. pHRI application

The sensors were integrated into each of the hands of the CASTOR robot. The robot's hand consists of a rigid 3D-printed cylinder covered by a soft fabric simulating the hand to provide a safe contact surface. The Velostat and FBG-based sensors were mounted in comparable internal positions to ensure similar boundary conditions. The robot was placed on a table facing the participant, with its arms in a neutral pose. Figure 3a illustrates the experimental setup, including sensor placement in the participant's view.

Participants were asked to perform a standardized handshaking interaction with the robot, as shown in Figure 3b. Before beginning, they were instructed to interact naturally but to maintain consistent, moderate force during the handshake. Each participant completed two interactions: one using the Velostat sensor and one using the FBG-based system. The interaction was free-time per trial, with ten trials for each sensor located in each robot's hands. The participant was guided to return to a neutral position after each handshake and wait 2 s for the next interaction. The sensors recorded the contact forces throughout the interaction.

IV. RESULTS

This section presents the results of the sensors' comparison, the single pixel calibration, and the behavior of both sensors, presenting the performance metrics, and for the pHRI evaluation, the force distribution in the robot's hand.

A. Single Pixel Calibration

Figures 4 and 5 show the calibration curves of the Velostat- and FBG-based tactile sensing matrices under controlled loading, highlighting distinct response behaviours when compared with the load cell (a–b for Velostat, c–d for FBG).

The Velostat sensor exhibited a nonlinear force–response relationship, well modelled by a quadratic regression over the 0–3 N range across the 36-pixel matrix. To complement the qualitative comparison, a statistical analysis of hysteresis was performed across the sensing elements of both matrices. Given the non-normal distribution of the data and the independent nature of the sensor groups, a Mann–Whitney U test was applied. The results did not indicate a statistically significant difference between the Velostat- and FBG-based sensors ($p = 0.13$). This result should be interpreted considering the unequal number of sensing elements in each matrix (36 versus 4) and the limited independence of pixel-wise measurements within each sensing system. Nevertheless, the Velostat sensor showed, on average, a lower hysteresis, and its effective sensitivity ranged from 0.74 at 0 N to -0.22 at 3 N (change factor -0.32).

In contrast, the FBG-based sensor exhibited a linear response over a wider force range and higher sensitivity in normalized $\Delta\lambda/N$, indicating greater responsiveness to small force variations. These results highlight a trade-off between linearity, sensing range, and hysteresis in hand-based pHRI: FBG sensors favour precision and linear force tracking, whereas Velostat sensors provide more stable responses under repeated interactions. Spatial resolution also differs, with 6×6 and 2×2 matrices for Velostat and FBG, respectively. Ground Sample Distance (GSD) defines the effective spatial resolution as the physical size of each sensing element. Table I summarises the technical performance of both systems.

B. pHRI evaluation

The heatmaps obtained during the handshakings with the five subjects are shown in Figure 6, representing the mean spatial distribution of the forces exerted on the robot's hand by each participant for both sensing systems. For each participant and sensor, ten handshaking trials were recorded, and the force values assigned to each sensing zone correspond to the average calibrated response across these trials.

Because the contact region produced by the participant's hand was larger than the spacing between adjacent sensing zones in the Velostat matrix, spatial interpolation was not required to represent the overall force distribution. The maps were therefore generated according to the native spatial layout of each sensor, with each heatmap cell corresponding directly to a physical sensing zone. This preserves the actual measured resolution of each system, avoiding the creation of artificial intermediate values. Thus, the Velostat heatmaps

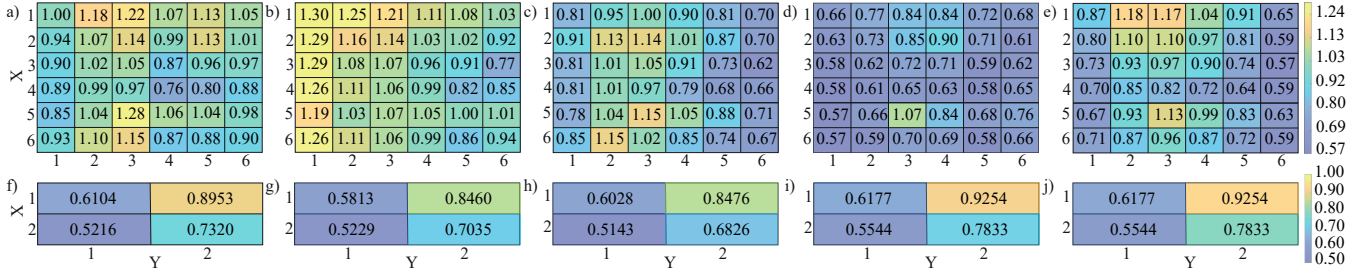


Fig. 6. Heatmaps for the mean distribution of force during the interactions maintained with the five subjects, A-E with the velostat matrix, and F-J with the FBG matrix.

TABLE I
SUMMARIZED RESULTS FOR BOTH SENSORS IN THE TECHNICAL COMPARISON.

Sensor	Equations	Order	Hysteresis	Range	Sensitivity	GSD _x (mm)	GSD _y (mm)	GSD (mm ²)	Pixel area (mm ²)	Fill factor (%)
Velostat	$y_u = -0.15x^2 + 0.70x + 0.19$ $y_d = -0.17x^2 + 0.78x + 0.20$	2	$6.10 \pm 1.27\%$	0-3N	Factor of 0.32	18.33	16.67	305.56	156.25	51.10
FBG	$y_u = 0.028x + 0.069$ $y_d = 0.027x + 0.073$	1	$7.14 \pm 1.13\%$	0-5N	0.027	50.00	75.00	3750.00	500.00	13.30

preserve the native 6 X 6 matrix resolution, whereas the FBG heatmaps preserve the native 2 X 2 sensing layout.

In both sensing systems, the heatmaps demonstrate that the sensors were able to capture regions of higher and lower contact during the interaction, reflecting the force pattern applied by the participants during the handshake. In the case of the Velostat-based sensor, the higher spatial resolution allows a more detailed identification of the force distribution over the surface of the robot's hand. For the five participants, the heatmaps exhibit well-defined local variations, highlighting differences among participants. In contrast, the heatmaps obtained with the FBG system present a more limited spatial representation. Although the system is capable of capturing the overall force distribution, it results in a spatially aggregated representation, which limits the identification of force variations across different contact locations.

V. DISCUSSION

Previous co-design studies in child-robot interaction have integrated tactile sensing primarily to detect predefined contact events such as hugging or holding [11], [34], or to recognise affective touch [35]. However, these systems are often implemented on custom or high-cost platforms, limiting their transferability to low-cost educational robots. In contrast, CASTOR was developed through participatory design with professionals as a low-cost modular robot for autism interventions [8], [9]. Within this context, this work extends existing approaches by providing a design-oriented comparison of two tactile sensing technologies integrated into the same robotic platform. Rather than focusing on touch classification, the study evaluates how each sensor represents spatial contact distribution and interaction variability, which were highlighted by teachers as relevant for interpreting engagement in educational settings. The results highlight key trade-offs between spatial resolution, force fidelity, robustness, and deployment constraints, supporting sensor selection

for child-robot interaction scenarios.

Velostat-based sensors showed characteristics consistent with prior studies, including nonlinear response, hysteresis, and inter-cell variability due to their polymeric structure [21]. Similar behaviour has been reported in related works, where calibration complexity is attributed to material properties [36]. In this study, a hysteresis of 6.10% was observed, comparable to values reported in the literature.

Although higher-resolution matrices (e.g., 27×27 and 32×32) have been proposed to improve spatial fidelity [19], [23], the 6×6 configuration used here provided sufficient resolution for distinguishing interaction patterns during handshakes. Crosstalk mitigation was achieved by setting non-selected rows to a high-impedance state, reducing undesired current paths. In contrast, the FBG-based system exhibited a predominantly linear response and high sensitivity, consistent with previous reports of FBG tactile skins [26]. While encapsulation may introduce hysteresis, this did not affect force estimation in the tested range. Overall, FBG sensors are better suited for precise force measurement, whereas Velostat-based sensors are more appropriate when spatial distribution and interaction pattern recognition are prioritised. Additionally, Velostat's polymeric structure supports repeated deformation, making it suitable for frequent physical interactions [22].

The Velostat sensor saturated at approximately 3 N per sensing zone but remained within its calibrated range during experiments. No saturation was observed in the recorded interactions, indicating no impact on heatmap interpretation. Considering the target application in child-robot interaction, such force levels are not expected to be exceeded in typical scenarios, although more variable interactions may occur in real deployments [12]. Therefore, future studies with children are required to validate these assumptions and establish application-specific force baselines. Finally, Velostat offers advantages in replicability and cost due to its fabrication and data acquisition [23]. In contrast, FBG sensors require complex inscription processes and optical interrogators, in-

creasing system cost and fragility, which may limit suitability for deployment in child–robot interaction contexts.

VI. CONCLUSION

This work presented the development and comparison of FBG- and Velostat-based tactile sensing matrices integrated into the hand of the CASTOR social robot. The results showed complementary characteristics between the two approaches. The FBG matrix provided higher sensitivity, a wider force-detection range, and a more linear response, whereas the Velostat matrix offered lower hysteresis, greater robustness, and substantially higher spatial resolution.

During hand-shaking interactions, both sensors successfully detected contact forces across the robot’s hand. However, the higher spatial resolution of the Velostat enabled more detailed localization of contact patterns, which is particularly important for touch interactions. Although the FBG sensor can measure a wider range of forces, the 3 N saturation limit of the Velostat sensor was not restrictive for the interaction scenarios considered in this study. Therefore, for tactile sensing in CASTOR’s intended child–robot interaction applications, the Velostat-based matrix represents the more suitable solution due to its spatial resolution, robustness, low cost, and ease of fabrication. Future work will evaluate the sensor with children and investigate real-time interaction classification and adaptive robot behaviours based on tactile feedback.

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